72	explore how science is used by society to make decisions, for example, the viability of solar cells as a replacement for other energy sources, the uses of remote sensing	

8.5 Digging up the Past (DIG)

The excavation of an archaeological site, from geophysical surveying to artefact analysis and dating:

- resistivity surveying
- artefact analysis by X-ray diffraction
- artefact analysis by electron microscopy.

Photons are used to model the behaviour of light, and waves to model electron behaviour.

There are opportunities to develop ICT skills using the internet and software simulations.

Through case studies, students learn how data can help resolve conflict and uncertainty, and how new knowledge is disseminated and validated.

There are opportunities for students to consider ethical issues concerning the digging of archaeological sites and removal of artefacts for scientific study.

Students will be assessed on their ability to:	**Suggested experiments**
57 investigate and use the relationship $R = \rho l/A$	Measure resisitivity of a metal and polythene
58 investigate and explain how the potential along a uniform current-carrying wire varies with the distance along it and how this variation can be made use of in a potential divider	Use a digital voltmeter to investigate 'output' of a potential divider
29 identify the different regions of the electromagnetic spectrum and describe some of their applications	
41 investigate and recall that waves can be diffracted and that substantial diffraction occurs when the size of the gap or obstacle is similar to the wavelength of the wave	Demonstration using a ripple tank

Students will be assessed on their ability to:	Suggested experiments
42 explain how diffraction experiments provide evidence for the wave nature of electrons	
48 explain how the amount of detail in a scan may be limited by the wavelength of the radiation or by the duration of pulses	
43 discuss how scientific ideas may change over time, for example, our ideas on the particle/wave nature of electrons	